

Metabolic and performance effects of raisins versus sports gel as pre-exercise feedings in cyclists.

Research suggests that pre-exercise sources of dietary carbohydrate with varying glycemic indexes may differentially affect metabolism and endurance. This study was designed to examine potential differences in metabolism and cycling performance after consumption of moderate glycemic raisins vs. a high glycemic commercial sports gel. Eight endurance-trained male ($n = 4$) and female ($n = 4$) cyclists 30 ± 5 years of age completed 2 trials in random order. Subjects were fed 1 g carbohydrate per kilogram body weight from either raisins or sports gel 45 minutes prior to exercise on a cycle ergometer at $70\% \dot{V}O_{2\max}$. After 45 minutes of submaximal exercise, subjects completed a 15-minute performance trial. Blood was collected prior to the exercise bout, as well as after the 45th minute of exercise, to determine serum concentrations of glucose, insulin, lactate, free fatty acids (FFAs), triglycerides, and beta-hydroxybutyrate. Performance was not different ($p > 0.05$) between the raisin (189.5 ± 69.9 kJ) and gel (188.0 ± 64.8 kJ) trials. Prior to exercise, serum concentrations of glucose and other fuel substrates did not differ between trials; however, insulin was higher ($p < 0.05$) for the gel (110.0 ± 70.4 $\mu\text{U} \times \text{mL}^{-1}$) vs. raisin trial (61.4 ± 37.4 $\mu\text{U} \times \text{mL}^{-1}$). After 45 minutes of exercise, insulin decreased to 14.2 ± 6.2 $\mu\text{U} \times \text{mL}^{-1}$ and 13.3 ± 18.9 $\mu\text{U} \times \text{mL}^{-1}$ for gel and raisin trials, respectively. The FFA concentration increased ($+0.2 \pm 0.1$ $\text{mmol} \times \text{L}^{-1}$) significantly ($p < 0.05$) during the raisin trial. Overall, minor differences in metabolism and no difference in performance were detected between the trials. Raisins appear to be a cost-effective source of carbohydrate for pre-exercise feeding in comparison to sports gel for short-term exercise bouts.